

CRCNS.org cai-3 data description

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Simultaneous two-photon imaging and electrophysiological recordings from neurons in the mouse retina and primary visual cortex using OGB and GCaMP6

Berens, P., Theis, L., Froudarakis, E., Reimer, J., Rosón, M. R., Baden, T., Euler T., Tolias, A.S. & Bethge, M.

Summary

This dataset provides data from N=50 neurons measured simultaneously using two-photon calcium imaging and electrophysiology in the mouse retina and visual cortex using different recording techniques, scanning rates and indicators. Importantly, all data has been acquired under typical population recording conditions in anesthetized and awake animals.

This dataset has been assembled for benchmarking spike inference algorithms:

Theis, L., Berens, P., Froudarakis, E., Reimer, J., Rosón, M. R., Baden, T., Euler T., Tolias, A.S. & Bethge, M. (2016). Benchmarking spike rate inference in population calcium imaging. *Neuron*, 90(3), 471-482. doi: <http://dx.doi.org/10.1016/j.neuron.2016.04.014>

In addition, parts of the data were used in the following publications:

Baden, T., Berens, P., Franke, K., Rosón, M. R., Bethge, M., & Euler, T. (2016). The functional diversity of retinal ganglion cells in the mouse. *Nature*, 529(7586), 345-350. doi:10.1038/nature16468

Reimer, J., Froudarakis, E., Cadwell, C. R., Yatsenko, D., Denfield, G. H., & Tolias, A. S. (2014). Pupil fluctuations track fast switching of cortical states during quiet wakefulness. *Neuron*, 84(2), 355-362. doi:10.1016/j.neuron.2014.09.033

Froudarakis, E., Berens, P., Ecker, A. S., Cotton, R. J., Sinz, F. H., Yatsenko, D., Saggau, P., Bethge, M. & Tolias, A. S. (2014). Population code in mouse V1 facilitates readout of natural scenes through increased sparseness. *Nature neuroscience*, 17(6), 851-857. doi:10.1038/nn.3707

Cotton, R. J., Froudarakis, E., Storer, P., Saggau, P., & Tolias, A. S. (2013). Three-dimensional mapping of microcircuit correlation structure. *Frontiers in neural circuits*, 7, 151. <http://dx.doi.org/10.3389/fncir.2013.00151>

Note that this is a subset of the data reported in Theis et al. (2016). The other subset has been set aside to serve as test set in a spike inference competition and will be

released once the competition is finished. The training set of the competition will be identical to the dataset released here.

Conditions for using the data

If you publish any work using the data, please cite the publication above (Theis et al. 2016) and also cite the data set in the following recommended format:

Berens, P., Theis, L., Froudarakis, E., Reimer, J., Rosón, M. R., Baden, T., Euler T., Tolias, A.S. & Bethge, M. (2016); Simultaneous two-photon imaging and electrophysiological recordings from neurons in the mouse retina and primary visual cortex using OGB and GCamp6. CRCNS.org.
<http://dx.doi.org/10.6080/K05D8PS8>

Methods

The methods used to acquire this dataset have been fully described in Theis et al. (2016).

We provide all imaging traces and spike trains resampled to a common resolution of 100 Hz to facilitate comparisons.

Data files organization

There is one file per dataset numbered as in the table below.

#	Area	Brain state	n	Indicator	Scan frequency	Scanning method	Field of view
1	V1	AN	11	OGB-1	322.5 ± 53.2	3D AOD	200x200 x100 μm^3
2	V1	AN	21	OGB-1	11.8 ± 0.9	2D galvo scan	250x250 μm^2
3	V1	AN	13*	GCamp6s	59.1	2D resonant	265x265 μm^2 135x135 μm^2
4	Retina	Ex vivo	6	OGB-1	7.8	2D galvo scan	100x100 μm^2
5	V1	AWK	9**	GCamp6s	59.1	2D resonant	265x265 μm^2

* 13 segments from 8 neurons

** 9 segments from 4 neurons

AN: in-vivo anesthetized, AWK: in-vivo awake

Each dataset is provided as a Matlab file (.mat) and a Python pickle file (.pck).

Data format

Matlab

The Matlab file contains a cell array with one cell per recording. This cell contains a struct with the following fields:

- calcium: calcium imaging trace
- spikes: binned spike train
- spike_times: exact spike times in ms
- fps: sampling rate (~100Hz)
- cell_num: only present in dataset 3 and 5; indicates which cell the segment was recorded from

The calcium trace and the spike train are provided at a sampling rate of 100 Hz, so each bin corresponds to 10 ms.

Example:

```
>> data{1}
```

```
ans =
```

```
    calcium: [1x71986 double]  
    spike_times: [1x2145 double]  
      spikes: [1x71986 uint16]  
        fps: 100.0002
```

Python

The data is available as a pickle-file. The data comes as an array of dictionaries with the same fields as defined above.

How to get started

Download the data, unpack it and run the included gettingstarted.m (Matlab) or open the gettingstarted.ipynb (Python). It will plot two example cells with calcium traces and spike rates.

How to get help

To get help with the data set post any questions on the forum at CRCNS.org.